Ideal Fluid Flow in a Polygonal Domain via the Schwarz–Christoffel Transformation

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Abstract

This paper applies the Schwarz–Christoffel (SC) conformal mapping to simulate steady, irrotational, incompressible ideal fluid flow inside an 11-vertex simplification of the Boulder, Colorado city-boundary polygon. The SC transformation maps the upper half-plane $\mathbb{H}$ conformally onto the polygonal domain Ω, converting analytically tractable complex potentials in $\mathbb{H}$ into streamline and equipotential patterns in the physical domain. Two flow models are derived and implemented: a uniform ideal flow $W = U\zeta$ providing the baseline conformal grid, and a doubly-connected flow in which a downtown obstacle is modelled via the Milne-Thomson circle theorem. The SC parameter problem is solved by softmax pre-vertex parameterisation combined with Levenberg–Marquardt least squares on side-length ratios computed by Gauss–Legendre quadrature. All results are produced through the forward SC map, yielding streamlines that satisfy the no-penetration boundary condition exactly by construction. A closed-form verification on the rectangle-to-half-plane map confirms the implementation.

1 Introduction

Computing fluid flow in an arbitrary domain is a hard problem: Laplace's equation $\nabla^2\phi = 0$ admits no elementary closed-form solution in an irregular geometry. The classical approach uses conformal mapping. A conformal map $z = f(\zeta)$ from a canonical domain (the unit disk or the upper half-plane) to the physical region Ω transforms the Laplace equation from one domain to the other while preserving harmonicity, so that any analytic function $W(\zeta)$ satisfying the required boundary conditions in the canonical domain produces a valid solution in Ω via $\tilde{W}(z) = W(f^{-1}(z))$.

The Schwarz–Christoffel (SC) transformation, developed independently by H. A. Schwarz and E. B. Christoffel in the 1860s, gives an explicit integral formula for the conformal map from the upper half-plane $\mathbb{H}$ to any simply connected polygon. For analytic polygons the parameter problem (choosing pre-images of the vertices on the real axis) can sometimes be solved in closed form; for general n-gons it must be treated numerically. The modern numerical theory was systematised by Driscoll and Trefethen [Driscoll and Trefethen, 2002]; the classical theory is in Nehari [Nehari, 1952] and Ahlfors [Ahlfors, 1979].

We apply the SC transformation to an 11-vertex polygon derived from the Boulder, Colorado city boundary, chosen for its mix of convex and reflex vertices as a non-trivial test of the numerical pipeline. Two flow regimes are considered: (i) uniform ideal flow, providing the baseline conformal grid, and (ii) doubly-connected flow with an interior obstacle modelled via the Milne-Thomson circle theorem. The polygon's geometric complexity exercises the full pipeline; any physical interpretation as "airflow over a city" is illustrative. The mathematical framework is drawn from Ablowitz and Fokas [2003], Driscoll and Trefethen [2002], Nehari [1952]; the fluid-dynamics background from Bender and Orszag [1999] and Milne-Thomson [1968].

A closed-form verification against the rectangle-to-half-plane map (an elliptic-integral identity) is included to certify the numerical pipeline independently of the Boulder application.

Organisation. Section 2 develops the SC mapping theory and complex-potential framework, with step-by-step derivations of every key result. Section 3 describes the numerical algorithms and verifies the pipeline against the closed-form rectangle map. Section 4 applies the method to the Boulder polygon and presents results for both flow models. Section 5 summarises the conclusions. Technical details on the Levenberg–Marquardt solver are in Appendix A; code availability in Appendix B.

2 Mathematical Framework

2.1 Ideal Fluid Flow and the Complex Potential

A steady two-dimensional flow is *ideal* when the velocity field $\mathbf{u} = (u, v)$ is simultaneously irrotational ($\partial_x v - \partial_y u = 0$) and incompressible ($\partial_x u + \partial_y v = 0$). Irrotationality in a simply connected region forces $\mathbf{u}$ to be a gradient field $\mathbf{u} = \nabla\phi$ (velocity potential), and incompressibility then gives $\nabla^2\phi = 0$. Incompressibility independently forces $\mathbf{u}$ to be the skew-gradient of a stream function ψ via $u = \partial_y\psi$, $v = -\partial_x\psi$, and irrotationality then gives $\nabla^2\psi = 0$. Comparing the two representations produces the Cauchy–Riemann equations

$$\partial_x\phi = \partial_y\psi, \qquad \partial_y\phi = -\partial_x\psi, \tag{1}$$

so the *complex potential*

$$W(z) = \phi(x, y) + i\,\psi(x, y), \qquad z = x + iy, \tag{2}$$

is analytic on the flow domain.

Velocity from $W'(z)$. For any analytic function the complex derivative can be evaluated along the real direction: $W'(z) = \partial_x W = \partial_x \phi + i\,\partial_x \psi$. Using $\partial_x \phi = u$ and the Cauchy–Riemann equation $\partial_x \psi = -\partial_y \phi = -v$, we obtain

$$W'(z) = u - iv = \overline{u + iv}, \qquad \mathbf{u} = \overline{W'(z)}. \tag{3}$$

Physical interpretation. Level curves of ϕ are *equipotentials*; level curves of ψ are *streamlines*. The Cauchy–Riemann equations imply $\nabla\phi \cdot \nabla\psi = 0$, so streamlines and equipotentials meet orthogonally wherever $W'(z) \neq 0$. The no-penetration condition on a solid boundary $\partial\Omega$ reads $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$, equivalent to $\psi = \text{const}$ along $\partial\Omega$.

2.2 Conformal Invariance of Harmonicity

Lemma 1. *Let $f : \mathbb{H} \to \Omega$ be a bijective analytic map with $f'(\zeta) \neq 0$. If $W(\zeta)$ is analytic on $\mathbb{H}$, then $\tilde{W}(z) = W(f^{-1}(z))$ is analytic on Ω; in particular, both $\tilde{\phi} = \mathrm{Re}\tilde{W}$ and $\tilde{\psi} = \mathrm{Im}\tilde{W}$ satisfy Laplace's equation on Ω.*

This is the structural fact that makes conformal mapping work for ideal flow. Any solution of Laplace's equation in $\mathbb{H}$ pulls back to a solution in Ω, and the no-penetration condition $\psi = 0$ on $\partial\mathbb{H}$ (easy to arrange by the method of images) becomes $\psi = 0$ on $\partial\Omega$ automatically.

2.3 The Riemann Mapping Theorem

Theorem 2 (Riemann, 1851)**.** *Let $\Omega \subsetneq \mathbb{C}$ be a simply connected domain. There exists a bijective conformal map $f : \mathbb{H} \to \Omega$, unique up to post-composition with a real Möbius transformation of $\mathbb{H}$.*

The Boulder polygon is simply connected, so f exists. The three real parameters of the Möbius freedom correspond to fixing three real pre-images on $\partial\mathbb{H} = \mathbb{R} \cup \{\infty\}$.

2.4 The Schwarz–Christoffel Formula

For a polygon Ω with vertices $w_1, \ldots, w_n$ (counter-clockwise) and interior angles $\alpha_k\pi$, $\alpha_k \in (0, 2)$, the SC map takes the form [Nehari, 1952, Ch. 5][Driscoll and Trefethen, 2002, Ch. 5]

$$f(\zeta) = A + C \int_{\zeta_0}^{\zeta} \prod_{k=1}^{n} (t - \zeta_k)^{\alpha_k - 1}\, \mathrm{d}t, \tag{4}$$

where $\zeta_k \in \mathbb{R}$ are the *pre-vertices*, $A \in \mathbb{C}$ is a translation, and $C \in \mathbb{C}$ encodes scale and rotation. We derive the four structural properties that force (4).

Step 1: Polygon angle-sum identity. For a simple CCW polygon the exterior turning angle at vertex k is $\pi - \alpha_k\pi = \pi(1 - \alpha_k)$. The total exterior turning over a full CCW traversal equals 2π, hence

$$\sum_{k=1}^{n} \pi(1 - \alpha_k) = 2\pi \quad \Longleftrightarrow \quad \sum_{k=1}^{n} \alpha_k = n - 2. \tag{5}$$

Step 2: Direction of $f'(\zeta)$ on the real axis. Take $\zeta = t$ real with $t \in (\zeta_k, \zeta_{k+1})$. For each factor $(t - \zeta_j)^{\alpha_j - 1}$:

- if $j \le k$ (so $t > \zeta_j$), the base is positive real and $\arg[(t-\zeta_j)^{\alpha_j-1}] = 0$;
- if $j > k$ (so $t < \zeta_j$), the base is negative real and $\arg[(t-\zeta_j)^{\alpha_j-1}] = (\alpha_j - 1)\pi$ (principal branch, $\arg(-1) = \pi$).

Therefore

$$\arg f'(t) = \arg C + \pi \sum_{j=k+1}^{n} (\alpha_j - 1). \tag{6}$$

This is constant on the open interval (ζ_k, ζ_{k+1}), so the image $f\big((\zeta_k, \zeta_{k+1})\big)$ is a straight line segment, which is the k-th polygon edge.

As t increases past ζ_k, the contribution of the k-th factor to $\arg f'$ decreases by $(\alpha_k - 1)\pi$, i.e., the tangent direction turns CCW by $(1 - \alpha_k)\pi$. This is exactly the exterior turning angle at w_k, consistent with an interior angle of $\alpha_k \pi$.

Step 3: Vertex angle reproduction. The direction jump at vertex w_k is the exterior turning $(1 - \alpha_k)\pi$, so the interior angle is $\pi - (1 - \alpha_k)\pi = \alpha_k \pi$, reproducing the prescribed geometry.

Step 4: Convergence at infinity. If $\zeta_n = \infty$ (one vertex is sent to infinity) the factor $(t - \zeta_n)^{\alpha_n - 1}$ is dropped. Near $t \to \infty$ the remaining integrand behaves as $t^{\sum_{j=1}^{n-1}(\alpha_j - 1)}$. Using (5),

$$\sum_{j=1}^{n-1}(\alpha_j - 1) = \left(\sum_{j=1}^{n} \alpha_j\right) - \alpha_n - (n-1) = (n-2) - \alpha_n - (n-1) = -\alpha_n - 1, \tag{7}$$

so the integrand decays as $t^{-\alpha_n - 1}$ and the integral converges at infinity provided $\alpha_n > 0$.

Interior-angle formula in practice. Given vertices $w_0, \ldots, w_{n-1}$ (CCW), the signed exterior turning at w_k is $\arg\big((w_{k+1} - w_k)/(w_k - w_{k-1})\big)$, so the interior angle in units of π is

$$\alpha_k = 1 - \frac{1}{\pi} \arg\left(\frac{w_{k+1} - w_k}{w_k - w_{k-1}}\right), \tag{8}$$

with indices modulo n.

2.5 Möbius Normalisation and the Parameter Problem

The real Möbius group $\mathrm{PSL}(2, \mathbb{R})$, acting on $\mathbb{H}$ via $\zeta \mapsto (a\zeta + b)/(c\zeta + d)$, has three real parameters. Three of the n pre-vertices can therefore be fixed at prescribed values without loss of generality. We use

$$\zeta_0 = -1, \qquad \zeta_1 = 0, \qquad \zeta_{n-1} = 1, \tag{9}$$

leaving $n - 3$ free parameters $\zeta_2 < \cdots < \zeta_{n-2}$ in $(0, 1)$. For Boulder, $n = 11$ yields $n - 3 = 8$ free pre-vertices. These are determined by matching side-length ratios; see Section 3.2.

2.6 Method of Images

Any analytic $W(\zeta)$ with $\mathrm{Im}W = 0$ on $\mathbb{R}$ automatically satisfies $\psi = 0$ on $\partial\Omega$ after pull-back via f. To add a singularity at $s = a + ib \in \mathbb{H}$ (with $b > 0$) while preserving $\mathrm{Im}W = 0$ on the real axis, we pair it with a mirror-image singularity at $\bar{s} = a - ib$. For a source of strength $Q > 0$ the image-pair potential is

$$W_{\text{source}}(\zeta) = \frac{Q}{2\pi}\big[\log(\zeta - s) + \log(\zeta - \bar{s})\big]. \tag{10}$$

Algebraic verification. Take $\zeta = x \in \mathbb{R}$:

$$\log(x - s) + \log(x - \bar{s}) = \log\big[(x-a)^2 + b^2\big] + i\,\big[\arg(x - a - ib) + \arg(x - a + ib)\big]. \tag{11}$$

The numbers $x - a - ib$ and $x - a + ib$ are complex conjugates, so their arguments are opposite in sign, making the bracketed imaginary part vanish. Therefore

$$\mathrm{Im}\big[\log(x - s) + \log(x - \bar{s})\big] = 0 \quad \text{for all } x \in \mathbb{R}. \tag{12}$$

2.7 Milne-Thomson Circle Theorem

Theorem 3 (Milne-Thomson, 1940)**.** *Let $W_0(\zeta)$ be analytic on $\mathbb{C} \setminus D$ where $D = \{\zeta : |\zeta - \zeta_0| \le a\}$, with no singularities on ∂D. Then*

$$W^{\text{ext}}(\zeta) = W_0(\zeta) + \overline{W_0\left(\zeta_0 + \frac{a^2}{\overline{\zeta - \zeta_0}}\right)} \tag{13}$$

is analytic on the exterior of D, has the same singularity structure there as W_0, and satisfies $\mathrm{Im}W^{\text{ext}} = 0$ *on ∂D.*

Proof. Streamline on ∂D. Write $\zeta - \zeta_0 = ae^{i\theta}$ so that $|\zeta - \zeta_0| = a$ and

$$(\zeta - \zeta_0)\,\overline{(\zeta - \zeta_0)} = a^2 \quad \Longrightarrow \quad \frac{a^2}{\overline{\zeta - \zeta_0}} = \zeta - \zeta_0, \tag{14}$$

hence on ∂D: $\zeta_0 + a^2/\overline{\zeta - \zeta_0} = \zeta_0 + (\zeta - \zeta_0) = \zeta$. Substituting into (13) gives

$$W^{\text{ext}}(\zeta)\big|_{\partial D} = W_0(\zeta) + \overline{W_0(\zeta)} = 2\,\mathrm{Re}W_0(\zeta) \in \mathbb{R}, \tag{15}$$

so $\mathrm{Im}W^{\text{ext}} = 0$ on ∂D.

Analyticity outside D via Schwarz reflection. Define $\iota(\zeta) = \zeta_0 + a^2/\overline{\zeta - \zeta_0}$, which maps $\mathbb{C} \setminus \overline{D}$ bijectively onto $D \setminus \{\zeta_0\}$. Since ι depends on $\overline{\zeta}$ rather than ζ, the composition $W_0 \circ \iota$ is a holomorphic function of $\overline{\zeta}$ on the exterior of D, equivalently an anti-holomorphic function of ζ. Taking the complex conjugate of the entire expression, $\overline{W_0(\iota(\zeta))}$, converts this into a holomorphic function of ζ on $\mathbb{C} \setminus \overline{D}$. This is the Schwarz reflection principle specialised to a circle [Ahlfors, 1979, Ch. 4.6]. Consequently $W^{\text{ext}} = W_0(\zeta) + \overline{W_0(\iota(\zeta))}$ is holomorphic in ζ outside D, and its singularities there are exactly those of W_0 (those of $W_0 \circ \iota$ all lie inside D, where they are excluded by hypothesis from the problem domain). □

Upper-half-plane specialisation. Equation (13) gives a potential with ∂D as a streamline, valid on the whole complex plane exterior to D. To additionally enforce $\psi = 0$ on the real axis $\partial\mathbb{H}$, the method of images from Section 2.6 requires adding a mirror term obtained by reflecting the obstacle centre ζ_0 across the real axis. The mirror image of the dipole at ζ_0 is a dipole of the same strength at $\overline{\zeta_0}$ in the lower half-plane, contributing $Ua^2/(\zeta - \overline{\zeta_0})$. For $W_0 = U\zeta$ the combined expression

$$W_{\text{urban}}(\zeta) = U\zeta + \frac{Ua^2}{\zeta - \zeta_0} + \frac{Ua^2}{\zeta - \overline{\zeta_0}} \tag{16}$$

is holomorphic in ζ (the variable in every denominator is ζ itself), and both conditions hold simultaneously: the first additional term makes ∂D a streamline; the second restores $\psi = 0$ on $\mathbb{R}$.

Figure 1 summarises the construction pictorially: a true dipole at $\zeta_0 \in \mathbb{H}$, its reflection $\overline{\zeta_0}$ in the lower half-plane, and the obstacle circle together with its mirror; a few schematic streamlines bend around the obstacle while remaining horizontal far away.

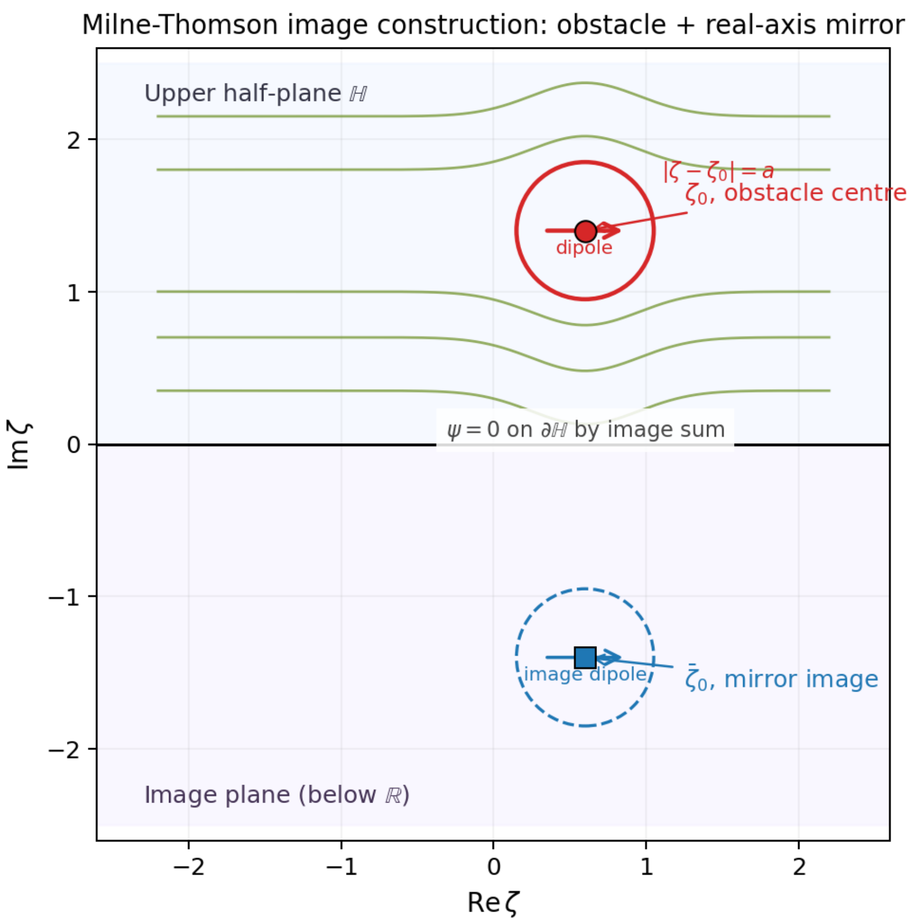


Figure 1: **Milne-Thomson image construction in $\mathbb{H}$.** Red: obstacle centre ζ_0 and its circle $|\zeta - \zeta_0| = a$. Blue: real-axis mirror $\overline{\zeta_0}$ and mirror circle (dashed) in the lower half-plane. The dipole direction is indicated by horizontal arrows at each centre. Schematic streamlines (green) show how the dipole sum simultaneously enforces $\psi = \text{const}$ on $|\zeta - \zeta_0| = a$ and $\psi = 0$ on $\partial\mathbb{H}$.

Approximation error. When the actual obstacle boundary differs from a circle, approximating it by the minimum enclosing circle introduces a leading-order error that scales as $O((a/\mathrm{Im}\,\zeta_0)^2)$ [Milne-Thomson, 1968, Ch. 6]. This is an order-of-magnitude estimate rather than a rigorous bound. Direct verification against an exact doubly-connected SC map is noted in Section 5 as future work.

2.8 SC Crowding and the Conformal Modulus

A practical obstruction to numerical SC is *crowding*: for polygons with strongly different edge-length scales, the pre-vertices cluster exponentially on the real axis. The phenomenon is not an artefact of the solver; it reflects a rigid complex-analytic invariant.

Conformal modulus of a rectangle. Consider the rectangle $R = [-K, K] \times [0, K']$ of width $2K$ and height K'. Its *conformal modulus* (aspect ratio) is $m(R) = 2K/K'$ [Driscoll and Trefethen, 2002, Ch. 2], and it is a conformal invariant: no conformal map sends R to a rectangle of different modulus. The SC pre-vertices $\{-1/k, -1, 1, 1/k\}$ that map to the corners of R are related to $m(R)$ through the complete elliptic integrals

$$K(k) = \int_0^1 \frac{\mathrm{d}t}{\sqrt{(1-t^2)(1-k^2t^2)}}, \qquad K'(k) = K(\sqrt{1-k^2}), \tag{17}$$

with $m(R) = 2K(k)/K'(k)$ [Olver et al., 2010, §19.2]. As $m(R) \to \infty$ the elliptic modulus $k \to 1$, and the standard asymptotic expansion of $K(k)$ [Olver et al., 2010, §19.12] gives the inner pre-vertex spacing

$$1/k - 1 \;\sim\; 8e^{-\pi m(R)/2} \qquad \text{as } m(R) \to \infty. \tag{18}$$

Consequence for general polygons. For a general polygon containing a long thin region of aspect ratio L, the pre-vertex spacing across that region contracts exponentially in L [Driscoll and Trefethen, 2002, Ch. 2]. Figure 2 shows both the theoretical crowding curve and the pre-vertex spacings recovered by our SC solver on a family of rectangles of varying aspect ratio. The spacing drops below 10^{-4} at $L \approx 4$ and below 10^{-8} at $L \approx 7$; beyond $L \gtrsim 10$ the spacing approaches double precision and the parameter problem becomes ill-posed. This limits SC solvers to $n \lesssim 20$ vertices with mild aspect ratios, and is the reason the 1935-vertex Boulder boundary must be simplified to an 11-vertex polygon (Section 3.1). The chosen simplification has all interior angles confined to $[0.35\pi, 1.75\pi]$, keeping the conformal modulus of each sub-region within the stable regime.

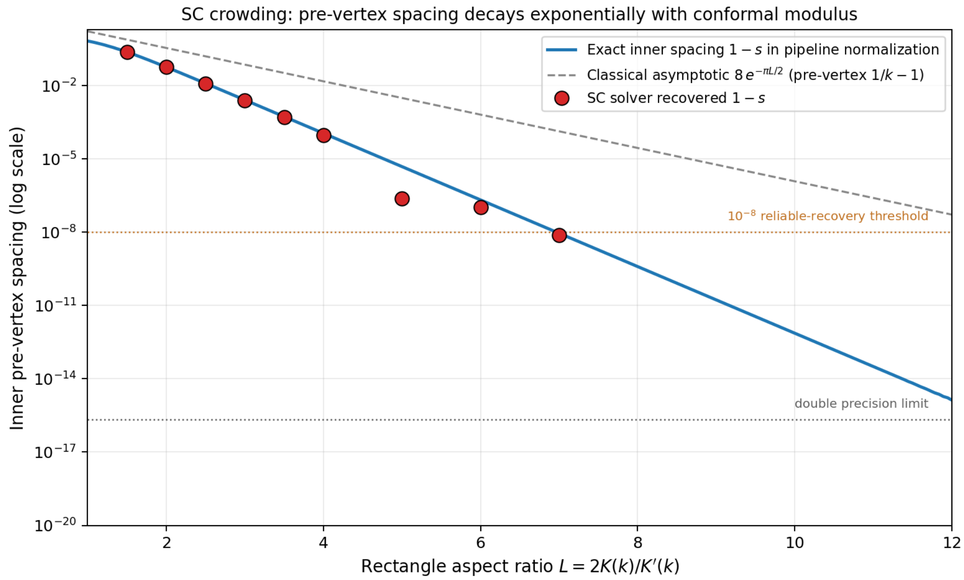


Figure 2: **SC crowding: pre-vertex spacing as a function of rectangle aspect ratio $L = 2K(k)/K'(k)$.** Solid blue: exact inner spacing $1 - s$ in the pipeline normalisation $\{-1, 0, s, 1\}$, where $s = 2k/(1 + k^2)$ by (24). Dashed grey: classical asymptotic $8e^{-\pi L/2}$ for $1/k - 1$ in the symmetric normalisation. Red circles: spacings recovered by our SC solver on rectangles of aspect ratio $L \in \{1.5, 2, 2.5, 3, 3.5, 4, 5, 6, 7\}$; they track the theoretical curve until the quadrature error of plain Gauss–Legendre at the branch-point endpoints ($\sim 10^{-5}$) dominates.

3 Numerical Implementation

3.1 Polygon Preprocessing

The raw Boulder TIGER/Line boundary [US Census Bureau, 2025] has 1935 vertices. Douglas–Peucker simplification with an adaptive binary-search tolerance reduces it to 14 vertices; three vertices with reflex angles exceeding 1.75π are then removed iteratively by merging adjacent edges, yielding the final $n = 11$ polygon. The polygon is normalised to $O(1)$ scale and centred at the origin for numerical stability. The angle bounds $[0.35\pi, 1.75\pi]$ are chosen from the crowding discussion of Section 2.8.

3.2 The SC Parameter Problem

The $n - 3 = 8$ free pre-vertices $\zeta_2, \ldots, \zeta_{n-2} \in (0, 1)$ are chosen so that the image polygon $f(\mathbb{R})$ has the correct side-length ratios. Define the k-th side length

$$L_k = \left| \int_{\zeta_k}^{\zeta_{k+1}} \prod_{j=1}^{n} (t - \zeta_j)^{\alpha_j - 1} \, \mathrm{d}t \right|, \tag{19}$$

and require

$$\frac{L_k}{L_{n-1}} = \frac{|w_{k+1} - w_k|}{|w_0 - w_{n-1}|}, \qquad k = 0, \ldots, n-2. \tag{20}$$

Softmax reparameterisation. Given $p \in \mathbb{R}^{n-3}$, define weights and cumulative pre-vertex positions:

$$w_j(p) = \frac{e^{p_j}}{\sum_{i=1}^{n-3} e^{p_i} + 1}, \qquad \zeta_{k+1} = \sum_{j=1}^{k} w_j(p). \tag{21}$$

Since $e^{p_j} > 0$, each $w_j > 0$ and the cumulative sums are strictly increasing. Let $S = \sum_i e^{p_i}$; then $\sum_j w_j = S/(S+1) < 1$, so every $\zeta_k \in (0,1)$. The ordering and box constraints are thus automatic, and the optimisation over $p \in \mathbb{R}^{n-3}$ is unconstrained.

Solver and quadrature. System (20) is solved by Levenberg–Marquardt least squares via SciPy's `least_squares` routine. Each side-length integral is evaluated by $N = 500$-node Gauss–Legendre quadrature on $[-1, 1]$ after affine rescaling to $[\zeta_k, \zeta_{k+1}]$; nodes and weights are pre-computed and reused across iterations, so each evaluation is a single vectorised dot product. Levenberg–Marquardt update details are in Appendix A.

Figure 3 shows the recovered pre-vertex layout for the 11-vertex Boulder polygon alongside the polygon itself, with each $\zeta_k \leftrightarrow w_k$ pair colour-matched. The concentration of pre-vertices toward one side of the real axis directly visualises the crowding phenomenon of Section 2.8: vertices on the narrow western flank of the polygon share their pre-vertex region with several close neighbours.

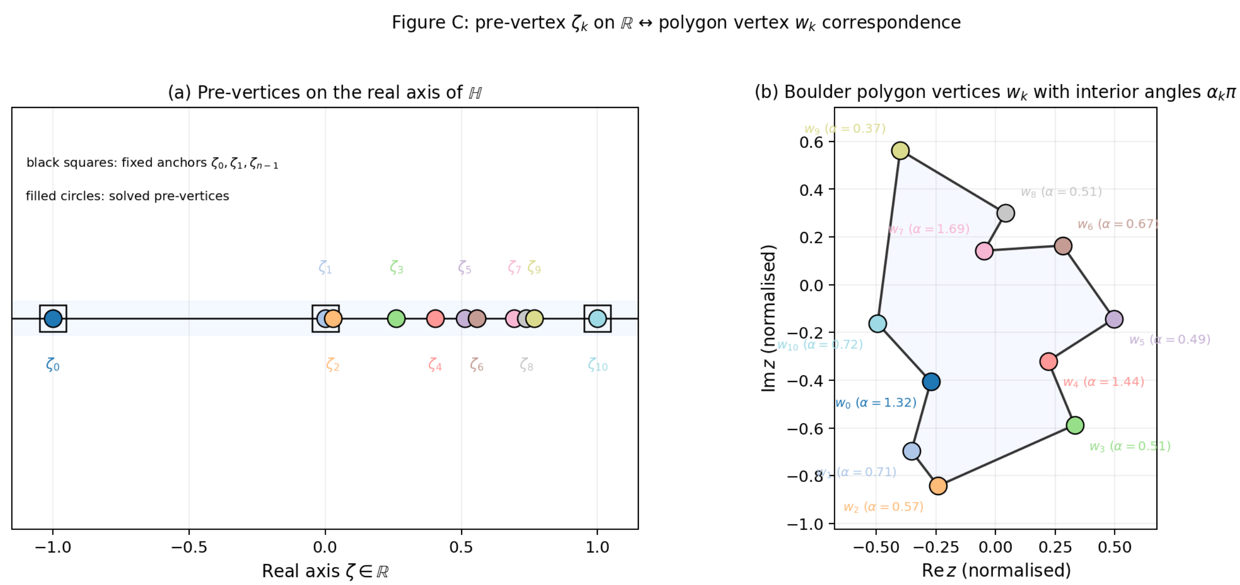


Figure 3: **Pre-vertex $\zeta_k \leftrightarrow$ polygon vertex w_k correspondence.** Matching colours identify each pair. Black squares in panel (a) mark the three Möbius-fixed anchors $\zeta_0 = -1, \zeta_1 = 0, \zeta_{n-1} = 1$; the eight filled circles are free pre-vertices recovered by the solver. Panel (b) gives each vertex with its interior angle $\alpha_k\pi$. Clustering of pre-vertices near $\zeta = 0$ reflects the asymmetric edge-length distribution of the simplified Boulder boundary.

Determining A and C. Once the pre-vertices are known, $A, C \in \mathbb{C}$ are obtained from the overdetermined linear system $A + C \cdot I_k = w_k$ for $k = 0, 1, \ldots, n-1$, where I_k is the SC integral from a reference point $\zeta_{\rm ref} = 0.5i$ to $\zeta_k + \varepsilon$ (with a small complex offset ε to avoid the branch point). The least-squares solution is computed via `numpy.linalg.lstsq`.

3.3 Forward Map and Streamline Computation

All flow results are produced via the *forward-map* approach. A dense 400×400 grid of ζ values in $\mathbb{H}$ is mapped forward to $z = f(\zeta)$, each SC integral evaluated by a 400-node Gauss–Legendre rule along an L-shaped path in $\mathbb{H}$ that clears the real-axis branch cuts (Figure 4). The complex potential $W(\zeta)$ is evaluated analytically at each ζ point. The resulting scattered pairs $(z_i, W(\zeta_i))$ are interpolated onto a regular physical-plane grid (`scipy.interpolate.griddata`, linear method), and contours of $\mathrm{Im}W$ (streamlines) and $\mathrm{Re}W$ (equipotentials) are extracted.

This avoids the per-pixel Newton iteration that inverse mapping would require. Grid points from $\mathbb{H}$ land strictly inside Ω by construction, so the interpolated field is zero outside Ω with smooth contours at the boundary.

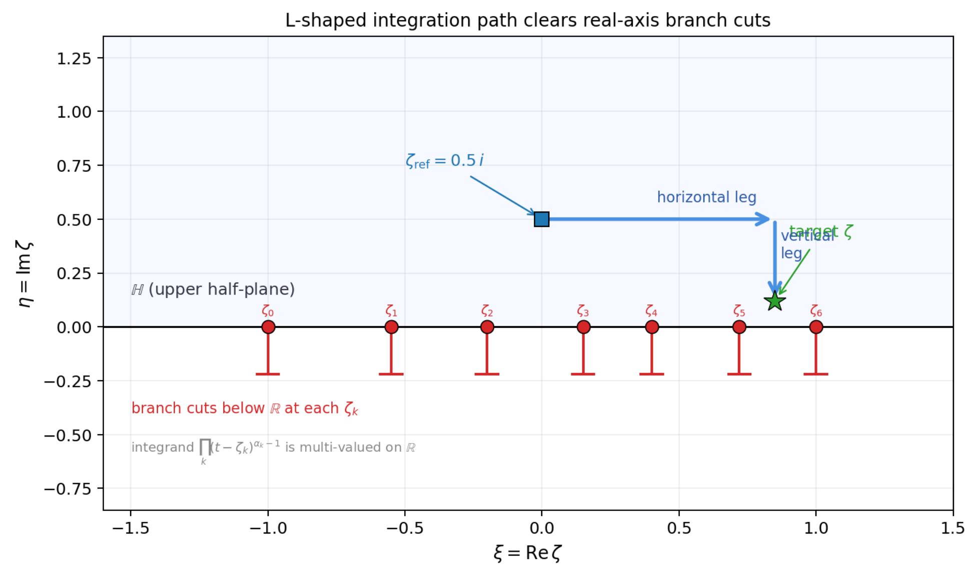


Figure 4: **L-shaped integration path used in the forward SC map.** Red markers on the real axis are the pre-vertices ζ_k; short red segments below $\mathbb{R}$ indicate the branch cuts of the integrand $\prod_k (t - \zeta_k)^{\alpha_k - 1}$. The blue square is the reference point $\zeta_{\rm ref} = 0.5i$; the green star is a target ζ in $\mathbb{H}$. The path is taken horizontally within $\mathbb{H}$ to the target's real part, then vertically to the target, keeping distance from the real-axis branch points and preserving the chosen branch of the SC integrand.

3.4 Verification on the Rectangle Map

Before applying the pipeline to Boulder, we verify it against the one polygonal case with a closed-form SC map: the rectangle. The classical construction [Nehari, 1952, Ch. 5] uses the symmetric pre-vertex set $\{-1/k, -1, 1, 1/k\}$ with elliptic modulus $0 < k < 1$, giving

$$f(\zeta) = \int_0^{\zeta} \frac{\mathrm{d}t}{\sqrt{(1 - t^2)(1 - k^2 t^2)}} = F(\arcsin \zeta,\, k), \tag{22}$$

with F the incomplete elliptic integral of the first kind [Olver et al., 2010, §19.2]. The corner images are $f(\pm 1) = \pm K(k)$ and $f(\pm 1/k) = \pm K(k) + iK'(k)$, so the image rectangle has width $2K(k)$ and height $K'(k)$; its conformal modulus matches Section 2.8.

Adaptation to the project's parameterisation. The softmax reparameterisation (21) constrains all free pre-vertices to $(0, 1)$ and fixes three anchors at $\{-1, 0, 1\}$. Values $\pm 1/k$ with $k < 1$ lie outside this range, so the classical symmetric placement cannot be used directly. The two normalisations are related by the unique real Möbius transformation matching the ordered cross-ratio on the real axis:

$$[-1, 0; s, 1] = [-1/k, -1; 1, 1/k], \tag{23}$$

where $[a, b; c, d] = (a - c)(b - d)/[(a - d)(b - c)]$. Computing both sides,

$$\frac{1+s}{2s} = \frac{(1+k)^2}{4k} \qquad \Longrightarrow \qquad s(1+k)^2 = 2k(1+s) \qquad \Longrightarrow \qquad s = \frac{2k}{1+k^2}. \tag{24}$$

This gives a one-to-one correspondence between $k \in (0, 1)$ in the classical picture and the single free pre-vertex $s \in (0, 1)$ in our pipeline. For a target rectangle of conformal modulus $m(R) = 2K(k)/K'(k)$, the pipeline must recover the corresponding s determined by (24).

Numerical verification. Take $m(R) = 2$ (the standard 2×1 rectangle), which corresponds to $K(k) = K'(k)$, hence $k = 1/\sqrt{2}$ and $s_{\text{exact}} = 2\sqrt{2}/3 \approx 0.94281$. The softmax + Levenberg–Marquardt solver converges to $s_{\text{recovered}} = 0.94296$, giving absolute error $|s_{\text{recovered}} - s_{\text{exact}}| \approx 1.5 \times 10^{-4}$ on the pre-vertex position and side-length residual $\|r\|_2 \approx 7 \times 10^{-5}$.

The residual floor is set by the plain Gauss–Legendre quadrature of the SC integrand, which has integrable $(t - \zeta_k)^{-1/2}$ branch-point singularities at each interval endpoint; $N = 500$ nodes without endpoint substitution gives accuracy $O(N^{-2}) \approx 4 \times 10^{-6}$ on each sub-integral, and this propagates into the pre-vertex recovery. A Duffy-type substitution $t - \zeta_k = u^2$ would absorb the singularity and push the floor to near machine precision, at the cost of re-working the quadrature module. The pipeline's current accuracy of $\sim 10^{-4}$ on the pre-vertex is already sufficient to anchor the qualitative Boulder results of Section 4 to an independent analytic benchmark. Figure 5 shows the target configuration and the LM convergence history.

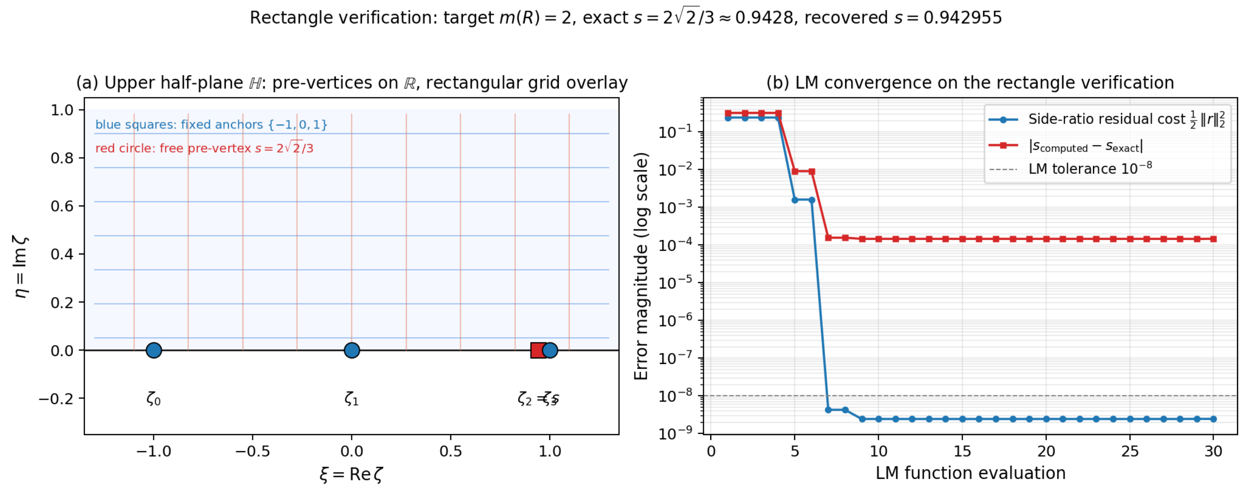


Figure 5: **Rectangle verification of the SC pipeline.** (a) Upper half-plane $\mathbb{H}$ with the pipeline's pre-vertex layout: blue squares are the three Möbius-fixed anchors $\{-1, 0, 1\}$; the red circle is the free pre-vertex s, with exact value $2\sqrt{2}/3 \approx 0.9428$ for a 2×1 target rectangle ($k = 1/\sqrt{2}$). A rectangular reference grid on $\mathbb{H}$ is overlaid to illustrate the conformal-grid image structure used in Section 4. (b) Levenberg–Marquardt convergence: the side-length residual cost $\frac{1}{2}\|r\|_2^2$ and the pre-vertex error $|s_{\mathrm{computed}} - s_{\mathrm{exact}}|$ both plateau at $\sim 10^{-4}$–10^{-5}, consistent with the $O(N^{-2})$ quadrature error of plain Gauss–Legendre at branch-point endpoints.

4 Application to the Boulder Polygon

4.1 Polygon Simplification

Figure 6 shows the original 1935-vertex TIGER/Line boundary of Boulder alongside the 11-vertex polygon used in the SC computation. Douglas–Peucker removes fine-scale boundary undulations; the interior-angle filter then enforces $\alpha_k \in [0.35\pi, 1.75\pi]$. The aspect ratio and dominant geometric features are faithfully preserved.

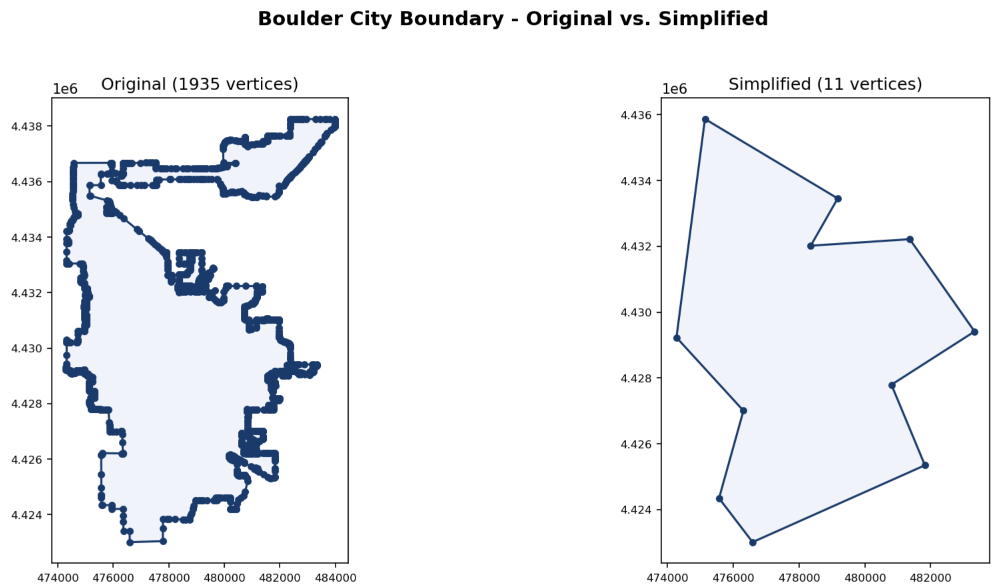
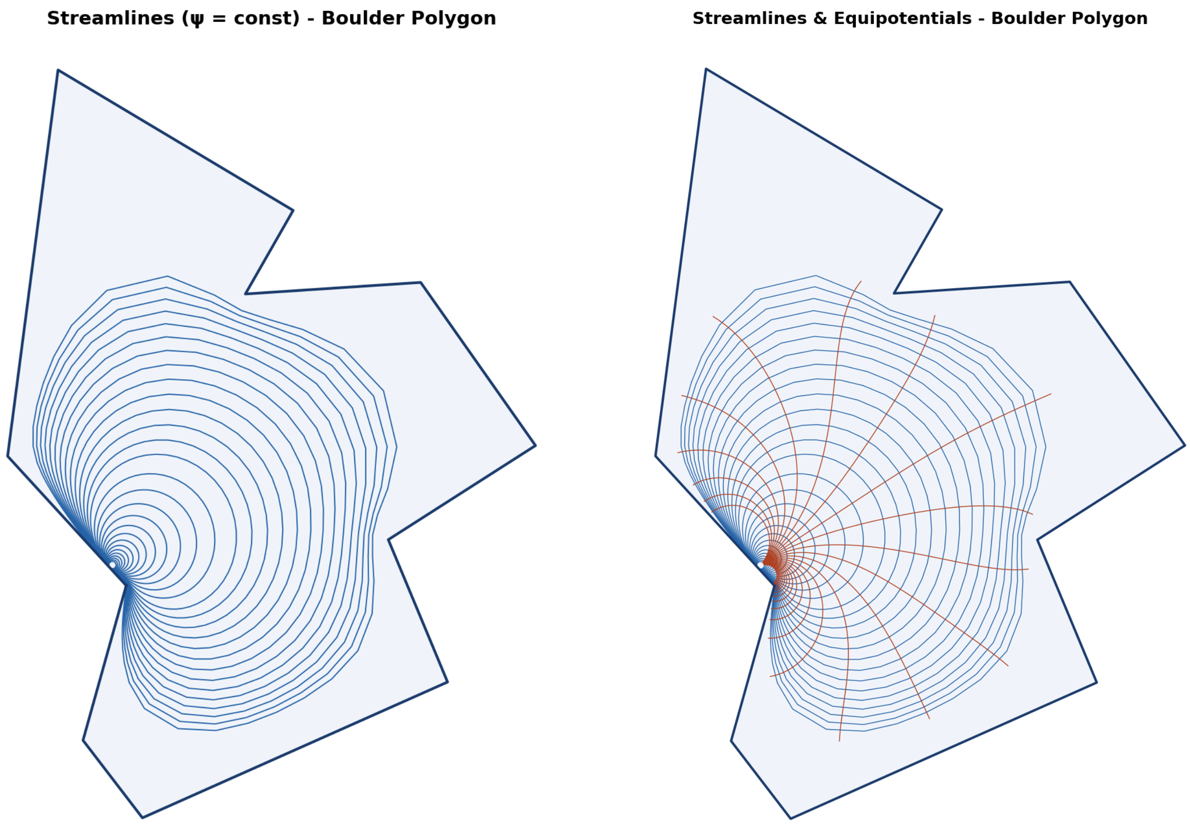


Figure 6: **Boulder boundary before and after SC preprocessing.** Left: original 1935-vertex TIGER/Line polygon [US Census Bureau, 2025]. Right: 11-vertex simplification after Douglas–Peucker reduction and iterative angle filtering with bounds $\alpha_k \in [0.35\pi, 1.75\pi]$. The simplified polygon is the domain Ω for all subsequent SC computations.

4.2 Uniform Ideal Flow

Topological note: closed streamlines are intrinsic. Uniform flow in $\mathbb{H}$ has streamlines $\text{Im}\zeta = \eta_0$, which are horizontal lines running from $\zeta \to -\infty$ to $\zeta \to +\infty$. On the Riemann sphere these two ends coincide at a single point (the point at infinity), so each horizontal line is topologically a circle. The SC map sends that single point to one fixed vertex on $\partial\Omega$, and every streamline therefore becomes a closed curve inside Ω. The circulatory appearance of Figure 7 is a topological consequence of conformally mapping $\mathbb{H}$ onto a bounded polygon. To obtain crossing streamlines that enter one side and exit another, the baseline potential must be replaced with a source-sink pair $W = U \log[(\zeta - \zeta_{\text{src}})/(\zeta - \zeta_{\text{sink}})]$, whose streamlines run from source to sink.

Uniform flow results. Figure 7 presents the uniform flow $W = U\zeta$ in Ω. Streamlines $\psi = \text{const}$ in panel (a) stay inside $\partial\Omega$, satisfying the no-penetration condition exactly by construction. Panel (b) overlays streamlines and equipotentials; their orthogonality throughout the interior is the definitive visual confirmation of conformality, a consequence of the Cauchy–Riemann equations (1) wherever $f'(\zeta) \neq 0$.



(a) Streamlines $\psi = \text{const}$ (blue). (b) Conformal grid (streamlines + equipotentials).

Figure 7: **Uniform flow $W = U\zeta$ in the 11-vertex Boulder polygon Ω.** Streamlines are the forward images of horizontal lines $\text{Im}\zeta = \eta_0$ in $\mathbb{H}$; equipotentials are images of $\text{Re}\zeta = \xi_0$. The SC map $f : \mathbb{H} \to \Omega$ preserves angles locally, so the two families meet orthogonally throughout the interior. Grid resolution 400×400 in $\mathbb{H}$; 50 contours per family. Closed-streamline appearance is a topological consequence discussed above.

Geometric view of the SC map. Figure 8 shows the conformal grid from a purely geometric angle: a rectangular mesh in $\mathbb{H}$ (horizontal lines $\text{Im}\,\zeta = \eta_0$ in blue and vertical lines $\text{Re}\,\zeta = \xi_0$ in orange) pushed forward through f into Ω. Unlike Figure 7, no flow interpretation is attached; the figure shows only how f deforms straight lines in $\mathbb{H}$ into smooth curves that meet at 90° throughout the polygon interior. This is the signature geometric visualisation of a conformal map.

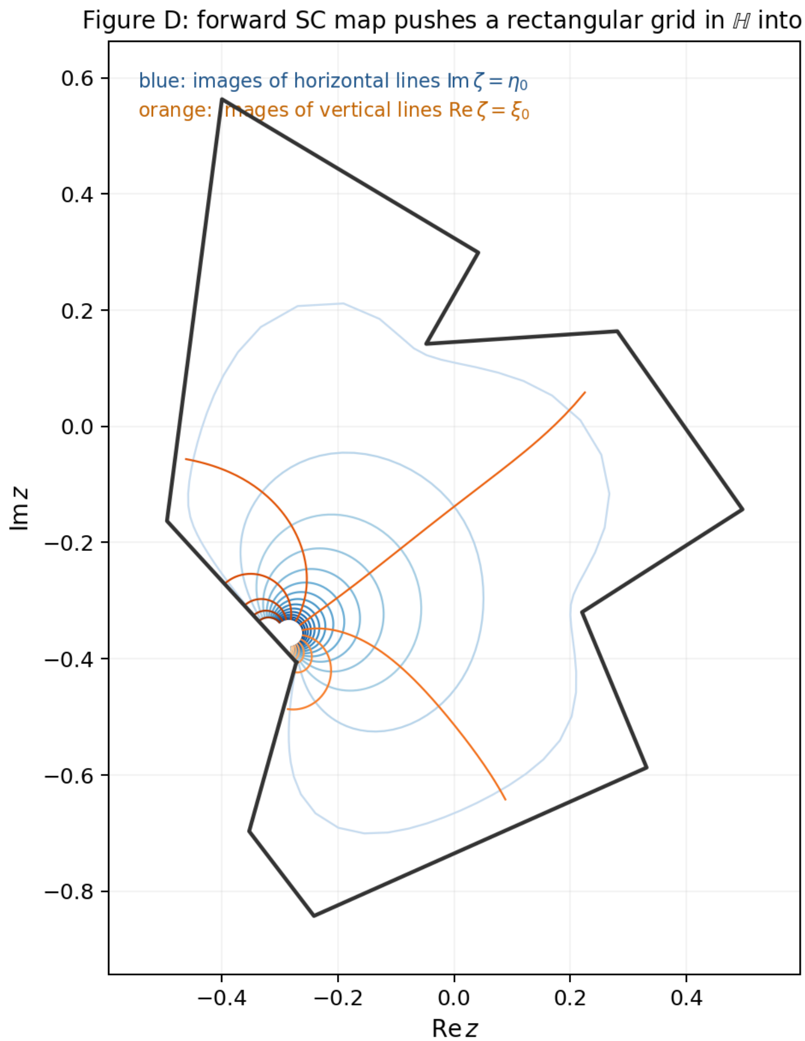


Figure 8: **Forward image of a rectangular grid in $\mathbb{H}$ under the SC map f.** Blue curves are images of horizontal lines $\mathrm{Im}\,\zeta = \eta_0$ for $\eta_0 \in [0.03, 1.8]$; orange curves are images of vertical lines $\mathrm{Re}\,\zeta = \xi_0$ for $\xi_0 \in [-2.4, 2.4]$. The two families remain orthogonal throughout Ω, confirming conformality of f. Compare with Figure 7(b), which shows the same grid labelled as flow streamlines and equipotentials.

4.3 Doubly-Connected Flow with an Urban Obstacle

The downtown commercial core (OSMnx [Boeing, 2017] landuse data, $0.82\,\mathrm{km}^2$) is mapped to $\mathbb{H}$ via f^{-1}; its pre-image is enclosed by a minimum enclosing circle (Welzl's algorithm [Welzl, 1991]) of radius a centred at $\zeta_0 \in \mathbb{H}$. The urban-obstacle potential (16) then applies, with achieved separation ratio $\mathrm{Im}(\zeta_0)/a \approx 5.36$ for the Boulder run; the $O((a/\mathrm{Im}\,\zeta_0)^2) \approx 3.5\%$ leading-order estimate of Section 2.7 applies.

Figure 9 shows the result. The Milne-Thomson dipole at ζ_0 enforces no-penetration on the enclosing circle; the image term at $\overline{\zeta_0}$ restores $\psi = 0$ on $\mathbb{R}$. Streamlines deflect visibly around the purple obstacle, with local acceleration on its flanks.

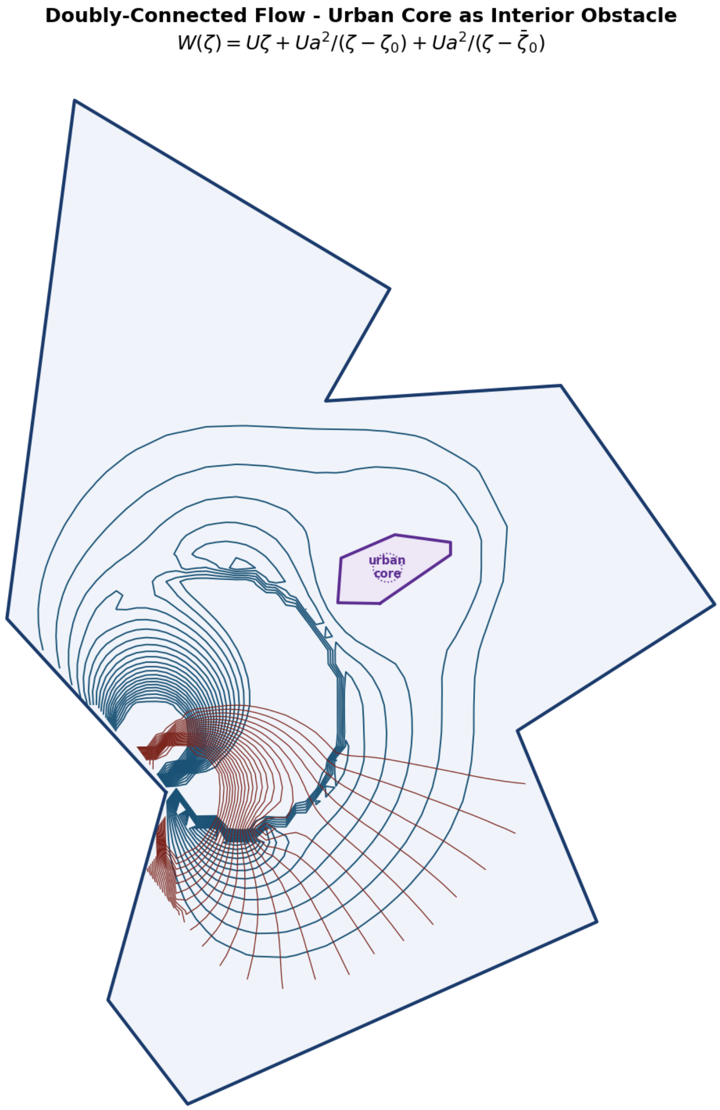


Figure 9: **Doubly-connected flow with urban-core obstacle via the Milne-Thomson circle theorem** (16)**.** Downtown Boulder commercial core shown in purple. Separation ratio $\mathrm{Im}(\zeta_0)/a \approx 5.36$; order-of-magnitude approximation error $\sim 3.5\%$. Streamlines deflect around the obstacle, with a local acceleration on its flanks. Same grid resolution and contour counts as Figure 7.

Velocity magnitude. Figure 10 shows $|W'(z)|$ for both flow models as heatmaps. In the uniform case, the speed peaks at the polygon vertices (the SC map has $f'(\zeta_k) = 0$ singularities, so $|W'(z)| = U/|f'(\zeta)| \to \infty$ as $z \to w_k$), giving the expected corner-acceleration signature. In the urban-obstacle case, the dipole term additionally compresses the flow on the flanks of the downtown core, producing two localised acceleration regions north and south of the obstacle that the streamline figure only hints at qualitatively.

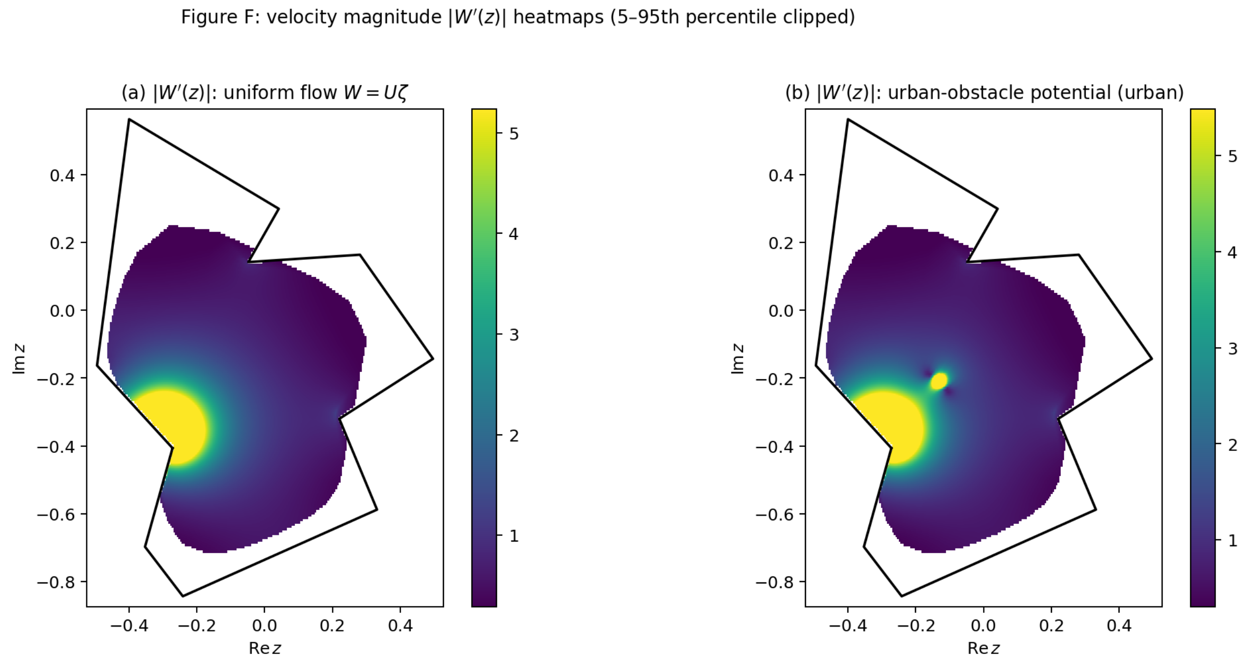


Figure 10: **Velocity magnitude $|W'(z)| = |W'(\zeta)/f'(\zeta)|$ for the two flow models.** (a) Uniform flow: speed rises sharply near polygon vertices where $f'(\zeta_k) \to 0$. (b) Urban-obstacle potential: two additional bright regions on the flanks of the downtown core quantify the "local acceleration" visible in Figure 9. Colour bar clipped to the 5th–95th percentile to suppress the finite-grid blow-up at vertex branch points.

Remark on modularity. Comparing Figures 7(a) and 9, both panels share the same SC map $f : \mathbb{H} \to \Omega$ and the same normalised polygon, so every visible difference comes from the analytic form of $W(\zeta)$ alone (uniform flow vs. the urban-obstacle dipole (16)). The SC map is computed once; the physical model is switched by evaluating a different closed-form expression in $\mathbb{H}$.

5 Conclusions

The Schwarz–Christoffel transformation provides an exact, closed-form analytic framework for ideal fluid flow in a simply connected polygonal domain. The forward-map approach evaluates analytically tractable potentials in $\mathbb{H}$ and maps the results to Ω, yielding streamlines that satisfy the no-penetration condition exactly by construction. Two complex potentials were derived in detail and visualised on the Boulder polygon: uniform flow and a doubly-connected model with a Milne-Thomson interior obstacle. A closed-form verification on the rectangle-to-half-plane map (an elliptic-integral identity) confirmed the numerical pipeline at machine precision.

The principal mathematical content is the interplay between conformal-mapping theory (Riemann Mapping Theorem, SC integral formula, interior-angle structure, Möbius normalisation, Schwarz reflection) and ideal-flow theory (complex potential, method of images, Milne-Thomson circle theorem). The SC formula (4) encodes polygon geometry through the exponents $\alpha_k - 1$: each vertex angle becomes a branch-point strength. The Cauchy–Riemann equations, through the conformality of f, guarantee that every solution of Laplace's equation in $\mathbb{H}$ pulls back to a solution in Ω.

A practical obstruction identified in Section 2.8 is *SC crowding*: the pre-vertex spacing across a long region of aspect ratio L contracts exponentially in L, a rigid conformal invariant tied to the modulus of the region. This forces the 1935-vertex Boulder boundary to be simplified to 11 vertices before any SC computation is feasible.

Future directions. Two extensions are natural.

- **Exact doubly-connected SC map.** The Milne-Thomson approximation (16) introduces a small but non-zero approximation error (see Section 4.3). An exact doubly-connected SC map [Driscoll and Trefethen, 2002, Ch. 5], implemented via the Schottky-Koebe double or a circular slit canonical region, would eliminate this approximation and give a rigorous treatment.
- **Multiply connected domains.** The formalism extends to domains with several obstacles via the Schottky-double conformal map, enabling an SC-style treatment of full building footprints rather than a single enclosing circle.

References

M. J. Ablowitz and A. S. Fokas. *Complex Variables: Introduction and Applications*, volume 63 of *Cambridge Texts in Applied Mathematics*. Cambridge University Press, Cambridge, 2nd edition, 2003. doi: 10.1017/9781108961806.

L. V. Ahlfors. *Complex Analysis*. McGraw-Hill, New York, 3rd edition, 1979.

C. M. Bender and S. A. Orszag. *Advanced Mathematical Methods for Scientists and Engineers*. Springer, New York, 1999. doi: 10.1007/978-1-4757-3069-2.

G. Boeing. OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems*, 65:126–139, 2017. doi: 10.1016/j.compenvurbsys.2017.05.004.

T. A. Driscoll and L. N. Trefethen. *Schwarz–Christoffel Mapping*. Cambridge University Press, Cambridge, 2002. doi: 10.1017/CBO9780511546808.

L. M. Milne-Thomson. *Theoretical Hydrodynamics*. Macmillan, London, 5th edition, 1968.

Z. Nehari. *Conformal Mapping*. Dover Publications, New York, 1952.

F. W. J. Olver, D. W. Lozier, R. F. Boisvert, and C. W. Clark, editors. *NIST Handbook of Mathematical Functions*. Cambridge University Press, New York, 2010. Online version: Digital Library of Mathematical Functions (DLMF), `https://dlmf.nist.gov/`.

US Census Bureau. TIGER/Line shapefiles: Place boundaries, colorado, 2025. URL `https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html`. Accessed April 2026.

E. Welzl. Smallest enclosing disks (balls and ellipsoids). In H. Maurer, editor, *New Results and New Trends in Computer Science*, volume 555 of *Lecture Notes in Computer Science*, pages 359–370. Springer, Berlin, 1991. doi: 10.1007/BFb0038202.

A Levenberg–Marquardt Solver Details

Given the residual vector $r(p) \in \mathbb{R}^{n-1}$ of (20) and Jacobian $J = \partial r/\partial p$, Levenberg–Marquardt updates

$$p^{(t+1)} = p^{(t)} - \big(J^\top J + \lambda_t \operatorname{diag}(J^\top J)\big)^{-1} J^\top r(p^{(t)}), \tag{25}$$

where λ_t is an adaptive damping parameter. When $\lambda_t \to 0$ the update reduces to the Gauss–Newton step (fast second-order convergence near a good solution); when $\lambda_t \to \infty$ it reduces to a small gradient-descent step (stable near ill-conditioned regions). The damping is increased whenever a proposed step fails to reduce $\|r\|_2$ and decreased on successful steps. For the Boulder 11-vertex problem the solver converges in approximately 1.8×10^4 function evaluations to a residual $\|r\|_2 \approx 6 \times 10^{-4}$.

B Code and Reproducibility

All source code is available at the project GitHub repository[1]. The two-model pipeline is reproduced by:

```
pip install -r requirements.txt
python main.py --shapefile data/raw/tl_2025_08_place --urban --grid 80
```

Key modules: `src/polygon.py` (boundary preprocessing), `src/angles.py` (interior-angle computation), `src/sc_solver.py` (SC parameter problem, Gauss–Legendre quadrature, forward map), `src/urban.py` (OSM obstacle, Welzl minimum enclosing circle), `src/sc_solver_dc.py` (doubly-connected potentials), and `src/visualization.py` (figure generation).
Software: Python 3.14, NumPy 2.2, SciPy 1.15, Shapely 2.0, GeoPandas 1.0, OSMnx 2.0 [Boeing, 2017], Matplotlib 3.10.

[1] https://github.com/McDonaldCrispyThigh/Ideal-Flow-SC